

hydro-graphical information like navigational hazards, depth of a particular place, width of the navigable channel and a lot more, as electronic data.

GPS is used to find out the precise position of the carrier and other ships. This positional information is fed to the ECDIS, which displays all relevant data so that the navigator gets the complete idea of his or her position and surroundings.

Radar AN/SPN-67, which is also used in a ship self-defence system (SSDS) is used for navigation. Along with this radar, a radar called AN/SPS-64 is also used for navigation. This radar operates on a fixed frequency of 9GHz and has a peak power output of 20kW. Its antenna can rotate at speeds up to 33rpm, which provides a high degree of target resolution and also allows target information update for every two seconds.

After detecting contacts through this radar, a system called automatic identification system (AIS) helps in identifying the nature of the contact. AIS is a system that all sea-going non-military vessels must possess. It relays its identification in radio frequency (like identify friend or foe, or IFF).



Fig. 33: Screen behind the woman sailor is ECDIS (Image courtesy: US Navy)



Fig. 34: A helmsman on the bridge of a carrier (Image courtesy: US Navy)

Generally, the required instructions for navigation are given by the captain from the combat direction centre (CDC). Necessary information like a copy of navigation radar displays is available at the bridge, which is the command centre for sailing the carrier. This bridge is the first storey above the flight deck. From here, sailors called helmsmen, control the sailing operations. Aegis escorts also have similar navigation radars and helmsmen controls.

Propulsion. A carrier's propulsion power comes from the electricity generated from two scaled-down nuclear reactors. These produce a shaft horsepower of 140MW to drive

propeller shafts. Electronic systems that are used in controlling the nuclear chain reaction and associated functions of normal reactors are also found in the carriers.

These reactors can work continuously for 20 years without requiring any refuelling.

With the aid of these reactors, carriers can reach speeds of 55kmph with ease. In naval parlance, sailing at this speed is like driving a Formula One car, and almost 100 thousand tonnes of mass sailing at this speed is something unimaginable.

Other vessels of the CSG are not nuclear powered but are powered by diesel-fuelled gas turbine engines. Controls of these propulsion systems also resemble industrial-class designs with military standard.

Russian carriers: The other side of the coin

Definitely, American CSGs are a force to reckon with. But one cannot forget Prince Hector or *Karna* while speaking about Achilles and *Arjuna*, respectively. So, such a detailed literature on American carriers will not be complete without mentioning the Russian carrier force.

Traditionally, Soviet Navy's and later Russian Navy's speciality is that these are a submarine dominant forces. Though there are very much capable surface vessels in their navies, dominance of their submarines is greater. As water is the best antidote for fire, submarines are a perfect match capable of standing toe to toe with the CSG. But it did not restrict Soviet Union to try its part on carriers.

Basically, pioneers in naval technology are Britons. British Empire